

Azizollah Taheri

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EDUCATION

PhD <i>Electrical and Computer Engineering at Northeastern University</i>	Expected 2027, Boston, US
MSc <i>Electrical and Computer Engineering at Northeastern University</i>	2025, Boston, US
BSc <i>Mechanical Engineering at Sharif University of Technology</i>	2021, Tehran, Iran

TECHNICAL SKILLS

Programming: Python, PyTorch, JAX, TensorFlow, OpenCV, MATLAB, C, C++, Julia, R
AI / ML: Reinforcement Learning, Machine Learning, Verifiable ML, LLMs, VLMs
Robotics: ROS, Gazebo, Webots, Motion Planning, Control Theory, Manipulation, Kinematics, Dynamics
Planning & Logic: Linear Temporal Logic (LTL), Signal Temporal Logic (STL), TWTL, PDDL
Platforms: TurtleBot, Warthog, Scout Mini, Crazyflie, DJI Tello, IRIS, Vicon, OptiTrack
Tools: Linux, Git, SolidWorks, SPSS

WORK EXPERIENCE

Academic Researcher	2022 – Present
- Developed a cooperative UAV–UGV planning algorithm using CLIP-based vision-language models (VLMs), enabling coordinated multi-robot decision-making in semantically unknown environments 2026, Boston, US - Designed a frontier-based temporal-logic motion-planning algorithm for unknown environments, eliminating stagnation with formal guarantees and boosting mission success rates by up to 65% 2025, Boston, US - Developed a formally verified motion-planning algorithm under temporal logic constraints for partially known environments, achieving 72% higher success rate and 3x faster planning — U.S. ARL funded 2025, Boston, US - Developed a time-window temporal-logic planning algorithm under contingency constraints, enabling robust task scheduling with formal guarantees — supported by the U.S. Army Research Laboratory 2024, Boston, US - Investigated RL frameworks for probabilistic spatio-temporal planning with time-window constraints, emphasizing robustness and constraint satisfaction — supported by DARPA 2023, Minneapolis, US - Led experimental studies on integrating state-estimation quality with temporal-logic compliance, reducing tracking error by 49% and measurement noise by 87% — MnDRIVE funded 2022, Minneapolis, US - Authored a bachelor's thesis analyzing the kinematics and dynamics of a 2-DOF pick-and-place robot and benchmarking it against two alternative configurations using MATLAB. 2021, Tehran, Iran Peer Reviewing: Reviewed papers for <i>IEEE RAL</i>, <i>IEEE IROS</i>, <i>IEEE ICRA</i> and <i>IEEE CDC</i>	

Teaching Assistant

• Statics, held office hours, graded homework and exams, University of Minnesota	2022, Minneapolis, US
• Aerospace Vehicle Design, mentored student projects, University of Minnesota	2022, Minneapolis, US
• Machine Learning, held office hours, graded homework and exams, Northeastern University	2026, Boston, US

PUBLICATIONS

- A. Taheri and D. Aksaray, “*Motion Planning Under Temporal Logic Specifications In Semantically Unknown Environments*”, American Control Conference (ACC), 2026
 - A. Taheri and D. Aksaray, “*Temporal-Logic-Aware Frontier-Based Exploration*”, American Control Conference (ACC), 2026
 - A. Taheri, Y. M. Phalle and D. Aksaray, “*Temporal Logic Motion Planning of a UAV-UGV Team in Semantically Unknown Environments*”, AIAA 2026 SciTech Forum
- Full publication list: [Google Scholar](#)